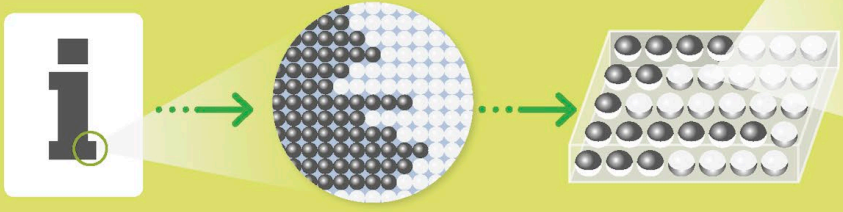
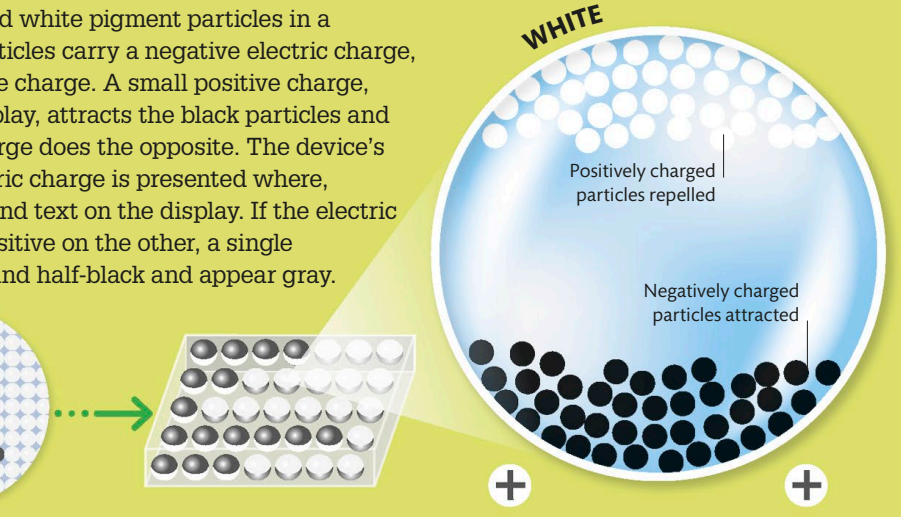
How electronic paper works

Inside electronic paper are thousands of tiny microcapsules, each containing black pigment particles and white pigment particles in a transparent, oily liquid. The black particles carry a negative electric charge, and the white particles have a positive charge. A small positive charge, provided by transistors under the display, attracts the black particles and repels the white ones. A negative charge does the opposite. The device's computer controls which kind of electric charge is presented where, building up black-and-white images and text on the display. If the electric charge is negative on one side and positive on the other, a single microcapsule will display half-white and half-black and appear gray.



10 X 100 X 200 X

Microcapsules
The microcapsules that make up text and images on electronic paper are each about the same width as a human hair.



KEY
⊕ Positive charge ⊖ Negative charge

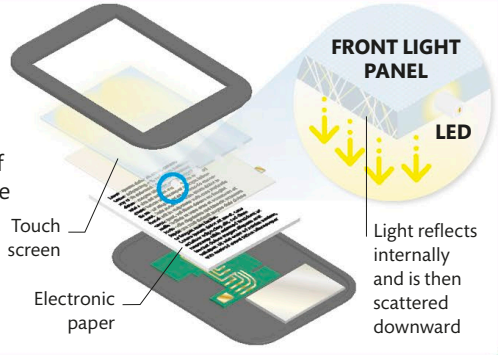
1 Black pigment particles are negatively charged, while the white particles are positive. A positive charge under the display attracts the black particles.

Electronic paper

Some e-readers display pages of text on screens made with electronic paper. Like real paper, electronic paper works with reflected light. This makes it better suited to reading text, because it causes less eye strain and works well in sunlight.

READING IN THE DARK

Electronic paper does not need its own light source, as computer screens do. For reading in the dark, however, many e-readers have LEDs along the side of the screen, to illuminate the display. The light travels across the inside of the transparent screen and is scattered downward onto the electronic paper.



IS IT BETTER TO READ AN E-PAPER TABLET BEFORE BED INSTEAD OF AN LCD TABLET?

It may be. Using a tablet can make it harder to fall asleep, because the blue light it emits can disrupt the action of the sleep-regulating hormone melatonin.

ELECTRONIC INK TECHNOLOGY IS BEING USED TO DEVELOP **CLOTHES** WITH **CHANGING PATTERNS**